Research Methods

Day 1: Introduction to Research & The Research Apprenticeship

Outline for Today

1. **Welcome & Bingo Icebreaker**
2. **What is Research? Opinion vs. Evidence**
3. **Global Challenges & Team Formation**

Welcome to **RESEARCH METHODS** by ResearchCORE!

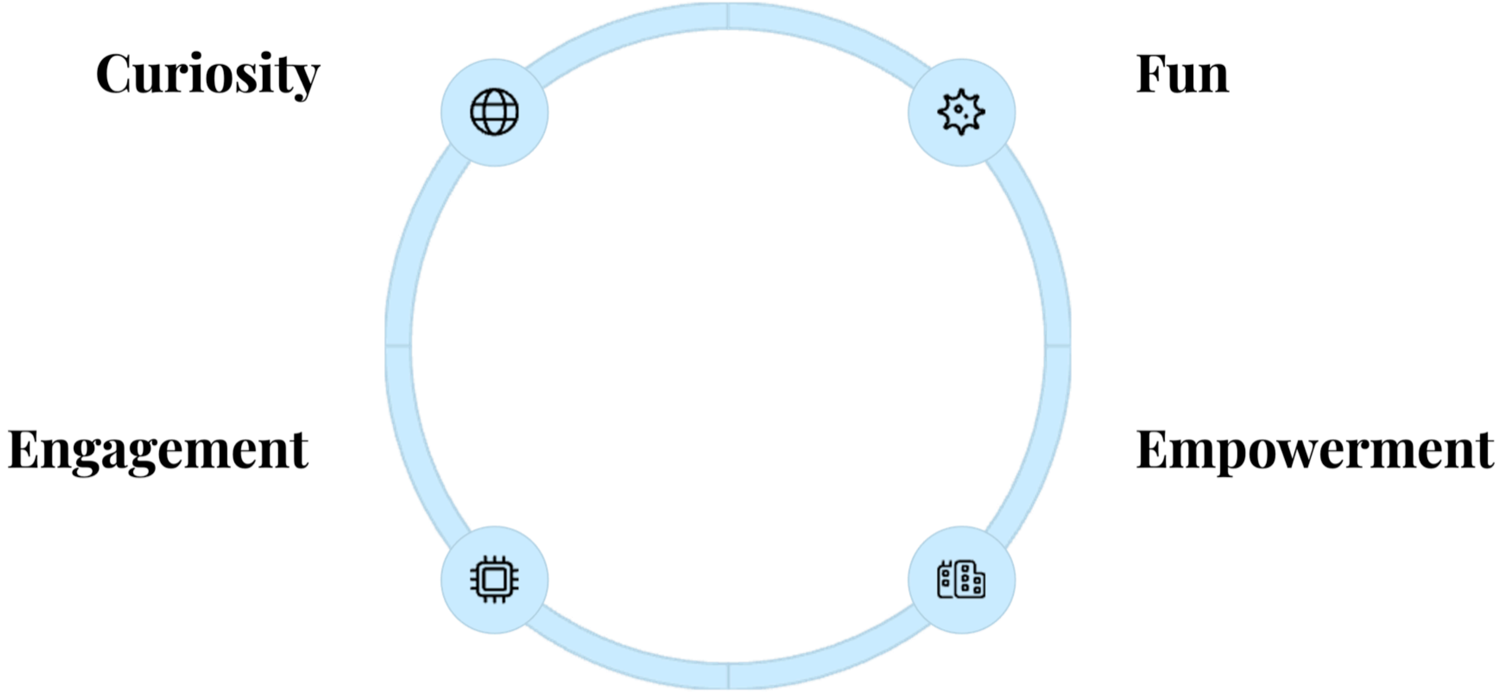
Hands-on, fast-paced experience where you will be researchers tackling real-world problems.

Journey Ahead: Your Very Own Research Process

- **Pick a Global Challenge**
- **Ask a Question**
- **Gather Information**
- **Analyze the Information**
- **Draw Conclusions**
- **Share Findings**

The final deliverables: A research proposal by Day 9 and a formal presentation by Day 10.

Our approach



Not only to learn how to conduct research, but to learn how to think like a researcher to

- Solve problems

- Challenge assumptions

- Make smart decisions

As you understand how the world works, you can change it for better!

Opinion vs. Research

The Difference Between Opinion and Research

Opinion

What you think or feel about something, based on personal perspective.

- **Subjective**
- Based on personal belief
- May lack supporting evidence

Research

Systematic investigation that produces evidence-based conclusions.

- **Objective**
- Based on data and evidence
- Follows methodical processes

Examples of Opinion

- 💬 Everyone should stop using social media completely.
- 💬 It's better to study late at night than early in the morning.
- 💬 Recycling doesn't make much of a difference to the planet.
- 💬 Fast food should be banned in school cafeterias.
- 💬 Climate change is too big of a problem for individuals to make a difference.

Examples of Research Statements

- 🧪 According to the **CDC's 2021 Youth Risk Behavior Survey**, which surveyed over 14,000 U.S. high school students, nearly **30% reported feeling persistently sad or hopeless** during the past year.
- 🧪 A 2020 global survey by **Pew Research Center** involving more than 13,000 adults in 13 countries found that 73% saw climate change as a major threat to their country.
- 🧪 A **randomized controlled sleep study at UC Berkeley** found that **teens who slept at least 8 hours** performed significantly better on memory tasks than those with less sleep. The study involved **105 adolescents aged 13–17**. *(Lopez et al., Sleep Research Journal, 2018)*
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Let's Play a Game:
MYTH or FINDING?

A **University of Chicago study** found that when people were forced to interact with strangers on their morning commute, they reported a significantly more positive experience than those told to sit in silence.

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FINDING

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👻 **MYTH**

Eating sugar causes children to become hyperactive and experience "sugar rushes" after consuming candy.

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⚗ **FINDING**

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⚗ **FINDING**

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MYTH

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FINDING

15:00

Let's take a 15 mins pre-assessment

Research

What is research?

Let’s explore the fundamental concepts of research, distinguish between opinions and evidence-based conclusions, and examine the research process across disciplines.

Systematic Research vs. Everyday Curiosity

Casual observation

Everyday noticing — informal, unplanned, and easy to be fooled by.

or

Systematic process

The same curiosity made rigorous — planned, repeatable, and built on evidence.

Same curiosity, two very different ways of answering it — **research** is the systematic one.

The Research Cycle

1. Ask a question

Start with a clear, focused question worth investigating.

2. Review the literature

Find out what's already known before adding to it.

3. Design the study

Plan a method that can actually answer the question.

4. Collect & analyze data

Gather evidence systematically and make sense of it.

5. Communicate findings

Share results so others can build on or challenge them.

Research is all around us

Not just labs — it happens in kitchens, gyms, classrooms, and on our phones.

- best study playlist?
- fastest route home?
- why did my bread flop?

What is Research?

A Journey of Discovery

Research is a systematic investigation to establish facts and reach new conclusions.

Question-Driven

Every great discovery begins with a well-framed question that guides your direction.

Evidence-Based

Research relies on gathering and analyzing evidence rather than opinions or assumptions.

Sports Analytics: Do Home Teams Really Win More?

Do home teams have a real advantage in the NBA — and if so, *why*?

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The Fan's Opinion

"Of course they do! The home crowd fires up the players and rattles the visitors."

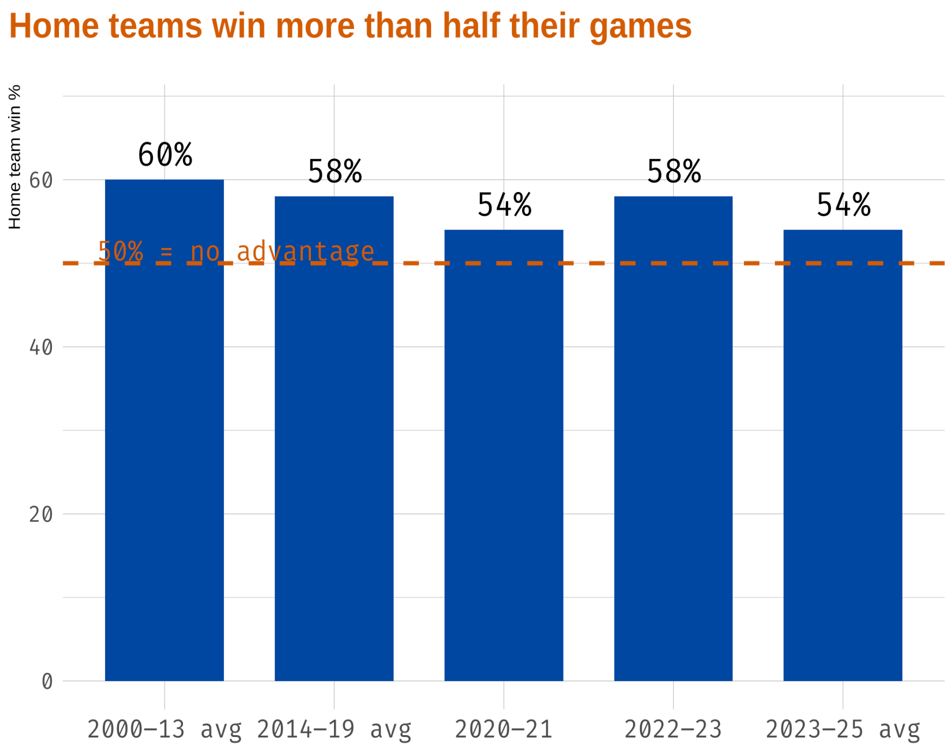
An Analyst's Hypothesis

Maybe it isn't the crowd at all — it could be travel fatigue, familiar rims, or favorable refereeing.

A Research Approach

Pull every NBA game from basketball-reference.com, compare home vs. away win rates, then test *which* story the evidence supports.

What the NBA Data Shows



Across two decades, NBA home teams have won **well above 50%** of regular-season games — but the edge has **shrunk over time**.

- **~60% home wins in 2000–2013**
- **Down to ~54% in recent seasons**
- **Lowest (54%) in 2020–21 — empty arenas**

So... is this a research question?

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"Home teams win more" is a *fact*. "*Why* do they win more?" is the research question.

What evidence would we need?

Crowd size, travel distance, rest days, referee calls — data that can separate the competing explanations.

What would 'prove' it?

We don't "prove" — we build evidence. The **empty-arena 2020–21 season** acts like a natural experiment on the crowd effect.

From Curiosity to Focused Questions

Curiosity Spark

The urge to know more about something that intrigues you.

Wonder

Asking yourself "Why is this happening?" or "How does this work?"

Research Question

Transforming curiosity into a clear, specific question to explore.

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WHAT ARE YOU CURIOUS ABOUT?

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A Key Attitude of a Researcher: Inquisition

- **"How do I know that?"**
- **"What information is this opinion based on?"**
- **"What's that claim based on?"**
- **"Has anyone actually tested it?"**
- **"Where is the evidence?"**

Why Evidence Matters

47%

Misinformation

Percentage of online content containing inaccurate claims.

73%

Better Decisions

Improved outcomes when using evidence-based approaches.

5X

Problem-Solving

More effective solutions when based on research.

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Case Study: How We Learned Smoking Causes Cancer

1

Observational data

Doctors noticed lung cancer climbing as cigarette use spread.

ask a question

2

Form a hypothesis

Could cigarettes be causing these cancers, not just coincide with them?

review the literature

3

Design the study

Compare smokers and non-smokers in large case-control and cohort studies.

design the study

4

Gather evidence

Smokers were far more likely to develop cancer — and more so the heavier the habit.

collect & analyze data

5

Drive policy change

Warning labels, ad bans, and public campaigns followed the findings.

communicate findings

Case Study: Research Behind a Taylor Swift Concert

1 Observational data

You notice the whole city seems happier and busier when Taylor Swift comes to town.

ask a question

2 Form a hypothesis

Maybe a huge concert lifts local mood — and spending — for a few days, not just coincidence.

review the literature

3 Design the study

Compare the town before, during, and after — survey mood, track hotels and restaurant sales.

design the study

4 Gather evidence

Packed hotels, busier restaurants, and surveys reporting higher happiness on concert days.

collect & analyze data

5 Share & inform decisions

The city plans better for big events, and businesses know when to staff up.

communicate findings

Small Group Activity: Research in Daily Life

Group Activity

15:00

Form 2 to 3 groups

3 to 4 members per group

Scenario 1: The Medical Trial

06:00

Dr. Lopez wants to know if a new blood pressure medication, **CardioPure**, is more effective than the current standard treatment. She tracks **200 patients** for six months. Half receive the standard drug, the other half receive CardioPure. At the end, she records the average drop in systolic blood pressure: the standard group dropped **8 mmHg**, while the CardioPure group dropped **14 mmHg** with no major side effects.

Research Question:	Data Collected:	Conclusion:

Scenario 2: Urban Planning

06:00

The **City of Springfield** wants to increase bike commuting and reduce traffic congestion, but only has budget to build **one** new protected bike lane. They consider two streets: **Oak Street** and **Maple Avenue**. For two weeks, electronic sensors count cyclists currently using both streets, and an online survey asks **1,000 residents** which route they prefer. The data: Maple Avenue has **300% more** daily cyclists than Oak Street, and **75%** of residents prefer Maple Avenue. The council votes to build the lane on Maple Avenue.

Research Question:	Data Collected:	Conclusion:

Scenario 3: Tech & E-Commerce

06:00

A popular clothing app, **TrendThread**, wants to see if changing their "Buy Now" button from blue to bright green will increase purchases. They run an **A/B test** for one week. Half of users randomly see the original blue button (**Group A**), the other half see the new green button (**Group B**). Analytics track clicks and completed checkouts. **Group A** had a **2.1%** checkout rate, while **Group B** had a **3.5%** checkout rate. The company permanently switches the button to green.

Research Question:

Data Collected:

Conclusion:

Scenario 4: Education & Teaching

06:00

Mr. Harrison wants to know whether interactive digital educational games help his 9th-grade algebra students understand quadratic equations better than traditional textbook worksheets. His **morning class** is taught using only the digital app; his **afternoon class** uses standard textbook worksheets. After two weeks, both classes take the **exact same** unit exam. The morning class (digital app) scores an average of **84%**, while the afternoon class (textbook) scores **76%**. Mr. Harrison decides to use the digital app in all his algebra classes next semester.

Research Question:

Data Collected:

Conclusion:

Scenario 5: Public Health Tracking

06:00

During a seasonal outbreak of a new COVID-19 variant, county health officials want to determine if the virus spreads faster in **indoor dining spaces** compared to **outdoor public areas**. They track positive case reports over a month, interviewing **500 infected residents** about recent locations, and collect **wastewater samples** to measure viral load. The interviews reveal **82%** of infected individuals had visited crowded indoor restaurants, compared to only **12%** who attended outdoor-only events. Wastewater data confirms massive spikes in areas dense with indoor venues. The government issues a temporary indoor mask advisory for restaurants.

Research Question:

Data Collected:

Conclusion:

GLOBAL CHALLENGES & TEAM FORMATION

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Climate Change

What it's about:

The planet you'll spend your whole life on is heating up. Burning fossil fuels traps heat, fueling the record heat waves, wildfire-smoke days, and flooding many of you have already lived through.

Key Facts:

- The global average temperature has already increased by **1.1°C (2°F)** since the late 1800s *(NASA, 2024)*.
- **2023 was the hottest year on record**, with ocean temperatures hitting record highs *(Copernicus Climate Change Service, 2024)*.
- Nearly **6 in 10 young people (ages 16–25)** say they are very or extremely worried about climate change *(Lancet, 2021)*.

Renewable Energy

What it's about:

The switch from fossil fuels to clean energy—solar, wind, hydro—is already showing up in your life: the EVs on your street, solar panels on rooftops, and how your future home and phone get powered.

Key Facts:

- In 2023, **30% of global electricity** came from renewable sources *(IEA)*.
- Solar power is now **cheaper than coal** in many countries *(BloombergNEF, 2023)*.
- **Nearly 1 in 5 new cars** sold worldwide in 2023 was electric—up from almost none a decade ago *(IEA, 2024)*.

Public Health Pandemics

What it's about:

You lived through one. COVID-19 closed schools, moved class to a screen, and canceled sports, concerts, and graduations—showing how fast a virus can upend everyday life.

Key Facts:

- COVID-19 led to over **770 million confirmed cases** globally by 2024 *(WHO)*.
- Global vaccine distribution exposed **serious inequalities**: only 26% of people in low-income countries received at least one vaccine dose by late 2022 *(Our World in Data)*.
- At the peak in 2020, schools closed for **about 1.6 billion students** across **190+ countries**—likely including yours *(UNESCO, 2020)*.

Poverty and Inequality

What it's about:

Two students with the same talent can face completely different futures based on family income—shaping whether they can afford a laptop, internet, tutoring, or college.

Key Facts:

- Over **700 million people**—about 1 in 11 globally—live on **less than $2.15 a day** *(World Bank, 2024)*.
- The **richest 1%** of the world's population owns almost **half of all global wealth** *(Oxfam, 2023)*.
- About **2.6 billion people are still offline**, making online schoolwork and opportunities far harder to reach *(ITU, 2023)*.

Educational Access

What it's about:

Imagine not being able to finish high school. For millions of teens, school isn't a given—it's blocked by cost, conflict, or simply being a girl.

Key Facts:

- **244 million children and youth** are out of school worldwide *(UNESCO, 2023)*.
- Girls are **more likely to be excluded** from school in low-income countries, especially during crises like war or pandemics.
- About **160 million children**—many of them teens—are in child labor instead of school *(ILO & UNICEF, 2021)*.

Digital Privacy

What it's about:

Every app you scroll—TikTok, Instagram, Snapchat—collects data about you: what you watch, where you go, and who you talk to. Who owns that data, and how it's used, is a growing fight.

Key Facts:

- US teens spend an average of **4.8 hours a day** on social media—nearly **6 hours** for 17-year-olds—every minute feeding data to apps *(Gallup, 2023)*.
- In 2023, **over 7 billion data records** were exposed in breaches *(IT Governance, 2024)*.
- Over **60% of teens don't understand** how their online data is collected, shared, or used *(Harvard Kennedy School, 2023)*.

Mental Health

What it's about:

The pressure you feel—school, grades, social media, the future—is part of a real and growing crisis in how young people are doing emotionally.

Key Facts:

- Globally, about **1 in 7 teens (ages 10–19)** lives with a mental health disorder *(WHO, 2024)*.
- In 2023, **about 40% of US high school students** felt persistently sad or hopeless *(CDC, 2023)*.
- Suicide is one of the **leading causes of death** for people aged 15–29 worldwide *(WHO, 2024)*.

Artificial Intelligence

What it's about:

From the feeds that pick your videos to chatbots that write essays, AI is reshaping how you learn, create, and—eventually—work. It also raises big questions about fairness and truth.

Key Facts:

- **ChatGPT reached 100 million users in about 2 months**—the fastest-growing consumer app at the time *(UBS / Reuters, 2023)*.
- **About 1 in 4 US teens** have used ChatGPT for schoolwork—**double** the share a year earlier *(Pew Research, 2025)*.
- The IMF estimates **~40% of jobs worldwide** are exposed to AI, reshaping the careers you're heading into *(IMF, 2024)*.

Misinformation

What it's about:

Half of what you scroll past could be misleading. Telling real news from rumors, ads, and AI-generated fakes is becoming a basic survival skill online.

Key Facts:

- On social media, **false news spreads about 6× faster** than the truth *(MIT, Science, 2018)*.
- Falsehoods are **70% more likely to be shared** than true stories *(MIT, Science, 2018)*.
- The World Economic Forum ranked **misinformation the #1 short-term global risk** for 2024, driven largely by AI *(WEF, 2024)*.

💧 Clean Water & Sanitation

What it's about:

You turn on a tap without thinking—but billions of people can't. Access to safe water and a working toilet still shapes health, school, and daily life around the world.

Key Facts:

- **2.2 billion people** still lack safely managed drinking water *(WHO/UNICEF, 2023)*.
- **3.5 billion people** lack safely managed sanitation, like a safe toilet *(WHO/UNICEF, 2023)*.
- In homes without water on tap, **women and girls** do most of the water-collecting—time that could be spent in school *(WHO/UNICEF, 2023)*.

What are you most curious about?

Write on a card: "I am most curious about _."

Share with others: Discuss your curiosity with classmates to spark ideas and collaboration.

Form teams of two or work individually if you prefer.

Looking Ahead: Tomorrow — Curiosity, Variables & Scientific Thinking

A **full day**: separating evidence from opinion, learning the building blocks of research, and meeting your team's real data.

Morning · ResearchCORE

- **Research or Opinion?** — a fast red-card / green-card game
- **Variables 101:** what you *change*, what you *measure*, and the sneaky *confounder*
- **Curiosities United:** sharpen your team's wonder into one focused question

Afternoon · DataCORE

- **Python basics:** storing variables & data types in Colab
- **Meet your dataset:** load & explore it with `pandas`
- **Cite it right:** define your IV & DV with a source

Homework

Handout 1: Introduction to Research

Finalize teams

Explore data-sets

Decide on your "Global Challenge"

- Climate Change
- Renewable Energy
- Public Health Pandemics
- Poverty and Inequality
- Educational Access
- Digital Privacy
- Mental Health
- Artificial Intelligence
- Misinformation
- Clean Water & Sanitation

Tonight: Keep the Momentum Going

After-Class Team Session

3:15 – 5:15 PM · with your TA / peer mentor — produce a real output before dinner. Not free time.

- **Team meeting (3:15):** agree on your global challenge by consensus — each member ranks their top 3, pick the highest total
- **Mission Sheet Part 1** together — compare answers, discuss differences
- **DataCORE:** each member runs the **NBA dataset** notebook and logs **2 observations**
- **Due before dinner:** team name, topic, and a one-sentence research-interest statement in the shared doc

Evening Work Period

7:00 – 9:00 PM · individual + partner work during study hours. TA on call.

- **Mission Sheet Part 2:** a **KWL table** for your chosen topic
- Re-read the **"What is Research?"** one-pager from class
- *(RcS)* write a **200-word** research-project pitch — question, data source, expected finding
- **DataCORE Journal:** add one NBA-dataset observation that **surprised you**

Scientific Storytelling: The Research Story Spine

15:00

Every good presentation tells a **story** — with a beginning, a middle, and an end.

Beginning

Why does this matter? What did we ask?

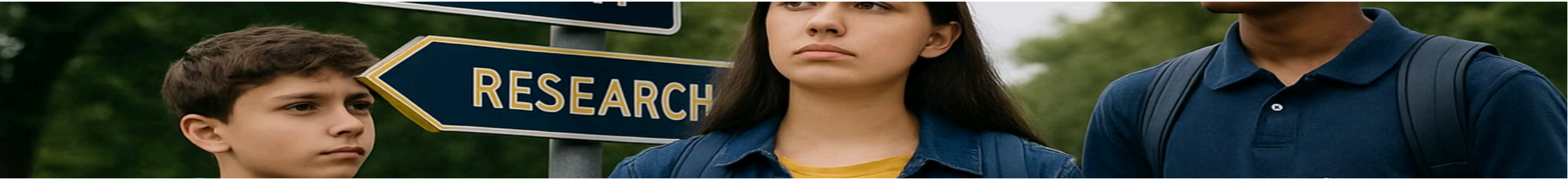
Middle

What did we do? What did we find?

End

What does it mean? What comes next?

In pairs: take one of your topic interests and tell a **60-second story spine** of a research project you'd want to do.



Let's wrap up this session!

Questions? Final thoughts?